

NRSG 741 Final Project - Milestone 2

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1 Project Information

GitHub Repository: NRSG 741 Project Repo

Data Source: Chayan, Ankur Ray (2024), Mendeley Data, V1, doi: 10.17632/8k9pvpmykk.1

2 Introduction

Maternal health disparities are persistent challenges across the globe. This project uses a dataset published by Chayan and Ankur Ray (2024), which includes a wide range of information about pregnant individuals—from basic demographics to key medical indicators. The dataset also classifies each pregnancy as either high-risk or not, based on clinical indicators like blood pressure, anemia, and fetal health. Additionally, the dataset contains variables related to socioeconomic status and prenatal care access. This analysis aims to explore these maternal health indicators and uncover demographic and clinical factors that influence maternal outcomes.

3 Data Loading

```
df <- read_excel("/Users/awulogejesseduncanmacclaud/Downloads/Book2\ 2.xlsx", sheet = "Sheet1")
```

```
## New names:
## * ' ' -> '...2'
## * ' ' -> '...3'
## * ' ' -> '...4'
## * ' ' -> '...5'
## * ' ' -> '...6'
## * ' ' -> '...7'
## * ' ' -> '...8'
## * ' ' -> '...9'
## * ' ' -> '...10'
## * ' ' -> '...11'
## * ' ' -> '...12'
## * ' ' -> '...13'
## * ' ' -> '...14'
## * ' ' -> '...15'
## * ' ' -> '...16'
## * ' ' -> '...17'
## * ' ' -> '...18'
```

```
head(df)
```

```
## # A tibble: 6 x 18
##   'ANCC REGISTER' ...2 ...3 ...4 ...5 ...6 ...7 ...8 ...9 ...10 ...11
##   <chr>           <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Name          Age  Gravida tt va~ gest~ weig~ heig~ bloo~ bloo~ jaun~ feta~
## 2 Rituporna     18   1st    1st   38 w~ 50 kg 5.3'' 100/~ None None Norm~
## 3 Moina         25   2nd    2nd   38 w~ 60 kg 5.2'' 100/~ None None Norm~
## 4 Rabeya        20   1st    1st   30 w~ 55 kg 5.0'' 100/~ None None Norm~
## 5 Shorna        22   1st    3rd   35 w~ 51 kg 5.4'' 110/~ None None Norm~
## 6 Tania Akter   20   1st    2nd   30 w~ 53 kg 5.2'' 100/~ None None Norm~
## # i 7 more variables: ...12 <chr>, ...13 <chr>, ...14 <chr>, ...15 <chr>,
## #   ...16 <chr>, ...17 <chr>, ...18 <chr>
```

4 Data Cleaning Step-by-Step

```
# 1. Set column names from the first row and remove the first row
colnames(df) <- df[1, ]
df <- df[-1, ]

# 2. Clean column names
df <- janitor::clean_names(df)

# 3. Basic renaming for clarity
names(df)[names(df) == "blood_pressure"] <- "blood_pressure_1"
names(df)[names(df) == "blood_pressure_2"] <- "blood_pressure_2"
```

```

# 4. Convert weight to numeric (remove 'kg')
df$weight_kg <- as.numeric(str_remove(df$weight, " kg"))

# 5. Convert height to numeric inches (remove '\')
df$height_in <- as.numeric(str_remove(df$height, "'"))

# 6. Extract gestational age (numeric weeks)
df$gestational_weeks <- as.numeric(str_extract(df$gestational_age, "\\d+"))

# 7. Extract systolic and diastolic BP from blood_pressure_1
df$systolic_bp <- as.numeric(str_extract(df$blood_pressure_1, "^\\d+"))
df$diastolic_bp <- as.numeric(str_extract(df$blood_pressure_1, "(?<=/)\\d+"))

# Check cleaned data
head(df %>% select(age, weight_kg, height_in, gestational_weeks, systolic_bp, diastolic_bp))

```

```

## # A tibble: 6 x 6
##   age  weight_kg height_in gestational_weeks systolic_bp diastolic_bp
##   <chr>      <dbl>      <dbl>          <dbl>          <dbl>          <dbl>
## 1 18          50        5.3             38            100            60
## 2 25          60        5.2             38            100            70
## 3 20          55         5              30            100            60
## 4 22          51        5.4             35            110            65
## 5 20          53        5.2             30            100            55
## 6 22          61        5.1             30            100            65

```

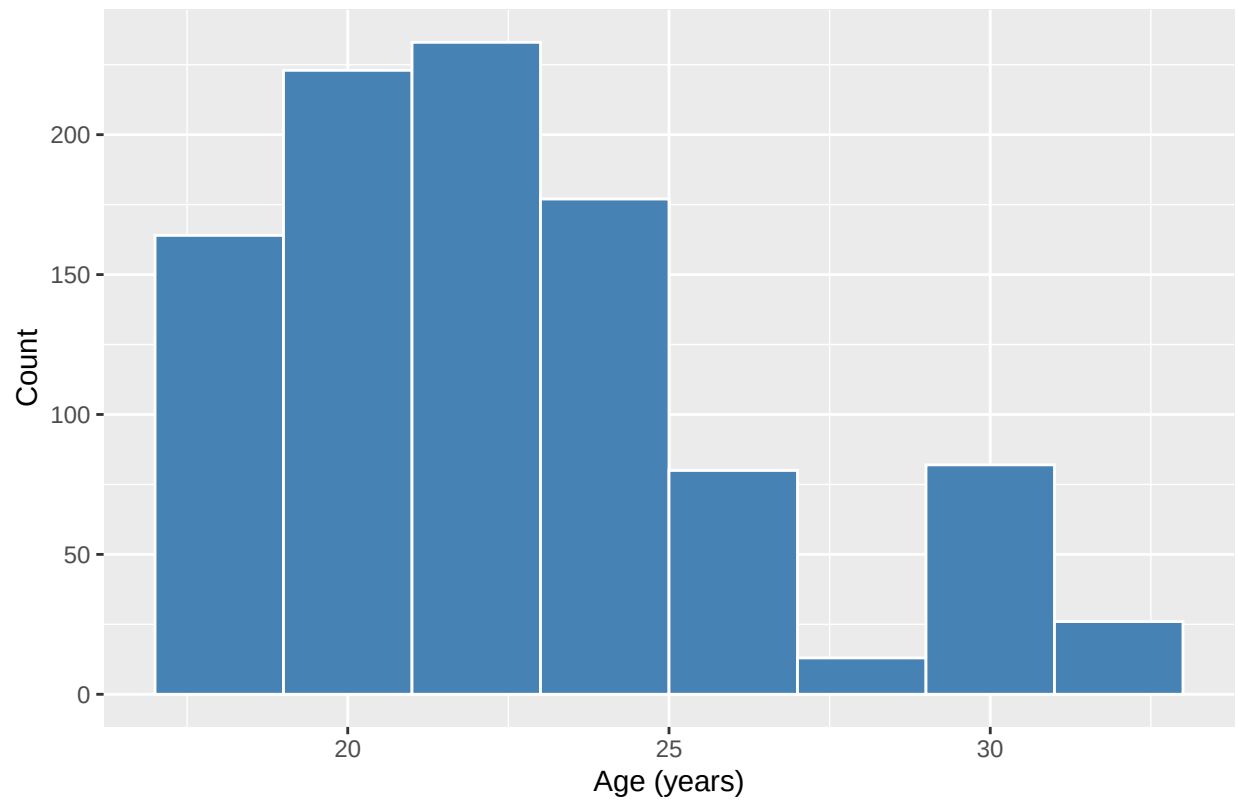
5 Exploratory Data Analysis

```

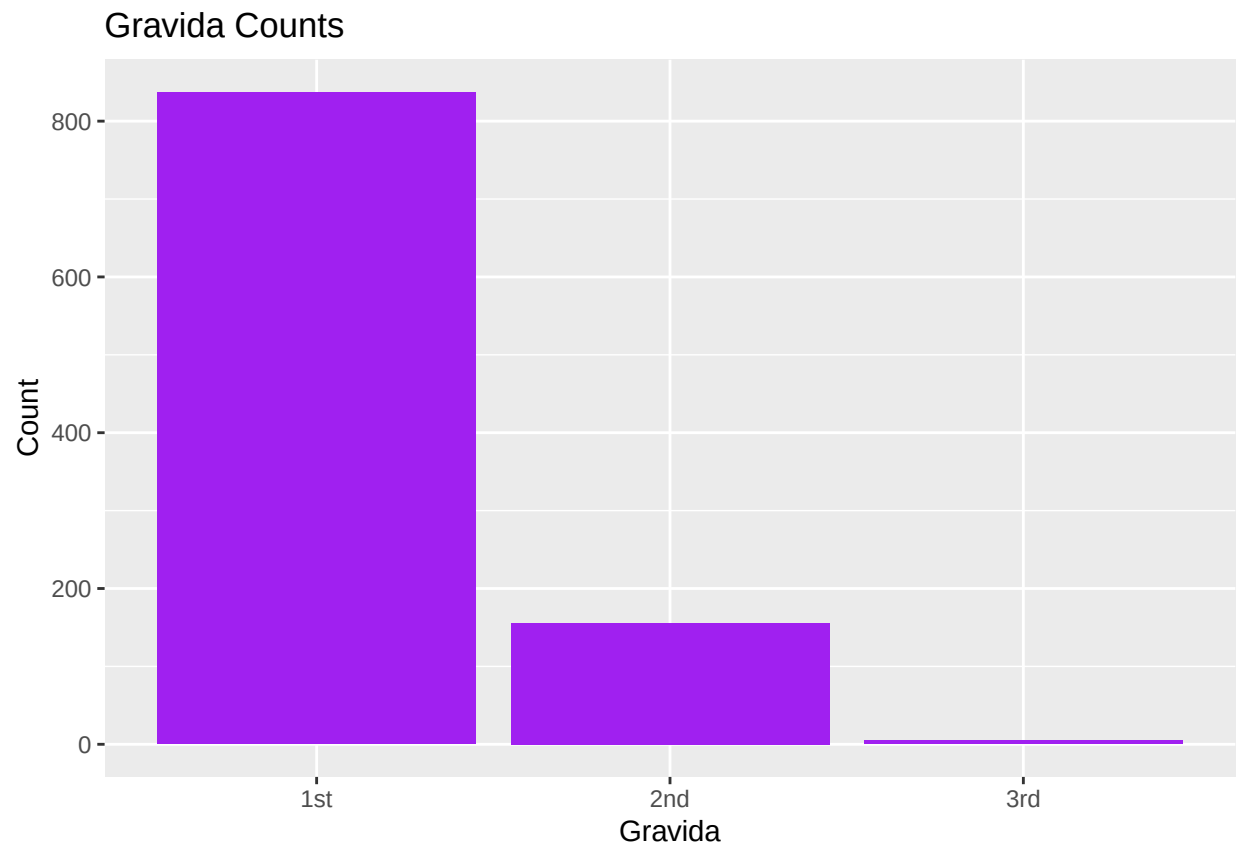
ggplot(df, aes(x = as.numeric(age))) +
  geom_histogram(binwidth = 2, fill = "steelblue", color = "white") +
  labs(title = "Age Distribution", x = "Age (years)", y = "Count")

```

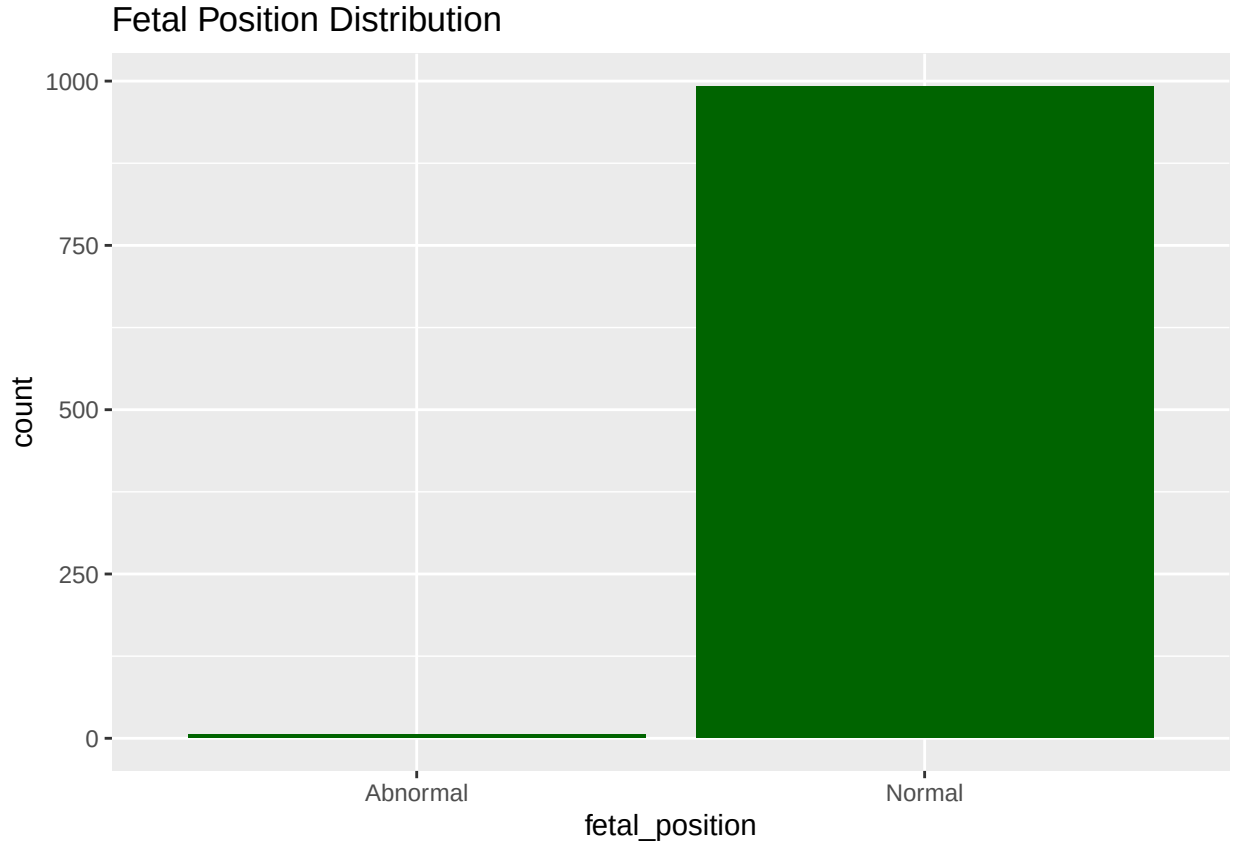
Age Distribution



```
ggplot(df, aes(x = gravida)) +  
  geom_bar(fill = "purple") +  
  labs(title = "Gravida Counts", x = "Gravida", y = "Count")
```



```
ggplot(df, aes(x = fetal_position)) +  
  geom_bar(fill = "darkgreen") +  
  labs(title = "Fetal Position Distribution")
```



6 Planned Analysis

plan to conduct the following:

- **Logistic regression:** Predicting high-risk pregnancy status using demographics and clinical indicators.
- **Linear regression:** Exploring predictors of fetal heartbeat.
- **Cluster analysis:** Identifying patterns in maternal profiles.
- **Machine learning:** Classification models to predict maternal health risks (if feasible).

7 Workflow Summary

- Data sourced from Mendeley Data (Chayan & Ankur Ray, 2024)
- Loaded and cleaned using a step-by-step pipeline
- Derived new variables (e.g., weight in kg, gestational weeks, systolic/diastolic BP)
- Performed preliminary visualization

8 Deviations from Plan

The original plan referenced data from Project Nell. However, the final dataset comes from the 2024 Mendeley dataset by Chayan and Ankur Ray, which was more comprehensive and publicly accessible.

9 Ancillary Questions

- Does systolic blood pressure or gravidity predict high-risk pregnancy?
- Are there patterns in fetal heartbeat related to prenatal care indicators?

10 Next Steps

- Finalize data cleaning and imputation
- Run regression and classification models
- Prepare final manuscript and presentation visuals